ROV Electrical Power Post-Mortem

Component-level voltage, current and power analysis across every operating case

Rev 1.0 · 10 November 2026 · 6S LiPo (22.2 V, 8 Ah) · 4- and 6-thruster configurations

Item	Value
Battery	6S LiPo, 22.2 V nominal, 8000 mAh, 177.6 Wh nominal / 142.1 Wh usable at 80 % depth of discharge
Operating bus window	25.2 V (full) → 19.8 V (cutoff, 3.30 V/cell); 18.0 V absolute floor
Hotel load (electronics, no lights)	5.7 W typical / 11.2 W peak on the 5 V rail; 47 mW on the 3.3 V sensor rail
Lighting	2 × 20 W at 12 V → 43.9 W from the battery at full brightness
Propulsion (per thruster)	815 W full throttle at 22.2 V (extrapolated) vs 400 W firmware cap
Peak battery current, all thrusters at cap	4T: 80.4 A | 6T: 124.1 A (before global budget)
Mixed-mission endurance	4T: 31 min | 6T: 21 min on 6.4 Ah usable
Analysis method	Embedded-systems workflow (constraints → architecture → drivers → validation → optimisation → test), rendered with the PDF skill

Contents

1. Executive summary
2. Scope, method and assumptions
3. Power architecture and tree
4. Battery analysis (source)
5. Component-by-component analysis (all 28 items)
6. Rail budgets and converter sizing
7. Propulsion deep-dive: thrusters on 6S
8. Operating-case study (C1–C8, 4T vs 6T)
9. Endurance and mission profile
10. Losses, voltage drop and thermal load
11. Protection coordination: fuses, wiring, contactor, pre-charge
12. Tether, Ethernet and emergency-stop loop
13. Firmware power management (embedded implementation)
14. Post-mortem findings register
15. Verification test plan
Appendix A. Formulas used
Appendix B. Data sources and confidence

1. Executive summary

This document is a full post-mortem of the ROV electrical design as defined by the final component list. Every one of the 28 line items has been assessed for supply voltage, current draw and power in each state it can be in (off, idle, active, peak, fault), then rolled up into rail budgets, battery current, voltage sag, losses, endurance and protection ratings. Two propulsion configurations (4 and 6 thrusters) are carried through every calculation.

The electronics are a small load — the whole computing, networking, camera and sensor stack draws about 6.3 W (0.28 A at 22.2 V). Lighting adds 44 W. Propulsion dominates everything: a single thruster can absorb more power than the rest of the vehicle combined, and the battery chemistry chosen (6S) pushes typical 20 V-class thrusters beyond their rating. The design is viable, but only with the corrections below.

Top findings (full register in Section 14)

F1 – Thruster over-voltage (Critical). A full 6S pack is 25.2 V; T200-class thrusters are rated to 20 V. Uncapped, a thruster at 25.2 V would draw ≈1083 W / 43 A, exceeding a 30 A ESC. A firmware power cap of 400 W per thruster (≈72 % throttle at full charge) is mandatory.
F2 – Main-bus over-current with 6 thrusters (High). Six thrusters at cap demand 124 A — above the 100 A fuse and XT90 rating. A global propulsion budget of 1700 W limits the bus to 86 A at nominal and 98 A at cutoff.
F3 – UNO Q USB host mode (High). Powering the UNO Q from VIN disables USB host, which would kill the camera and the Ethernet hub. Power it through the USB-C hub/PD input or the 5 V pin from the 5 V buck, and bench-verify.
F4 – Hardwired E-stop vs Gigabit (High). 1000BASE-T uses all four tether pairs, leaving no conductors for the surface E-stop loop. Run the link at 100BASE-TX and use pairs 4/5 + 7/8 for the stop loop (loop resistance ≈3.3 Ω).
F5 – Contactor inrush (Medium). Closing onto ≈3.0 mF of ESC capacitance with 18.8 mΩ source resistance gives a theoretical 1338 A spike. Add a 10 Ω pre-charge path (τ = 30 ms).

Metric	4-thruster	6-thruster
Armed idle (C2)	13.4 W / 0.60 A	15.6 W / 0.70 A
Inspection hover, lights on (C4)	98 W / 4.4 A	107 W / 4.8 A
Cruise (C5)	252 W / 11.4 A	427 W / 19.5 A
All thrusters at cap (C6)	1663 W / 80.4 A	2466 W / 124.1 A → budgeted 86 A
Mixed mission average	263 W / 12.3 A	375 W / 18.0 A
Mixed mission endurance	31 min	21 min

2. Scope, method and assumptions

The analysis follows the embedded-systems engineering workflow: (1) analyse constraints — voltage windows, current limits, power budget; (2) design the architecture — power tree, switching, protection; (3) define driver-level behaviour — sensor modes, converter operating points; (4) validate — derating checks against ratings; (5) optimise — losses and endurance; (6) define tests that confirm the numbers on hardware. Where a component on the list is generic (e.g. "Brushless DC Thruster Motors"), a representative part was selected and the assumption is stated so it can be swapped for the real datasheet value.

Key assumptions

Parameter	Value used	Basis / confidence
Thruster	Blue Robotics T200-class: 205 W @ 12 V, 390 W @ 16 V, 645 W @ 20 V full throttle, rated 7–20 V	Manufacturer data — High. Extrapolation above 20 V — Medium
Thruster power vs voltage	$P \propto V^{2.24}$ fitted to the three published points	Curve fit — Medium
Thruster power vs throttle	$P(u) = P_{max} \cdot u^3$ (propeller law)	Engineering approximation — Medium; verify on bench
ESC	Bidirectional BLHeli-type, ≈30 A continuous, 7–26 V, 50 mA idle, ≈4 % conversion loss	Typical for T200 ESCs — Medium
Battery	6S, 8 Ah, 25C (200 A) rating, 2.5 mΩ/cell → 15 mΩ pack	C-rating and IR to be read from the label/measured — Medium
Usable capacity	80 % DoD = 6.4 Ah	Design choice for cycle life
Distribution resistance	3.84 mΩ (connector, fuse, contactor, shunt, 8 AWG leads, PDB)	Estimated — Medium
5 V buck efficiency	90 %	Typical synchronous buck at 1–2 A
12 V buck efficiency	93 %	Typical at ~3.4 A from 22 V
LED lamps / drivers	20 W at 12 V each; drivers are PWM dimmers with ≈0.4 W loss at 100 %	Lamp rating given; driver type assumed
Contactor coil	24 V nominal coil, 3 W (R ≈ 192 Ω), pick-up ≤ 18 V	Must be confirmed on the chosen part
UNO Q	5 V 3 A max input; 3.5 W typical / 7.5 W peak used	Rated input — High; consumption estimate — Medium
Thruster layout	4T: 2 horizontal + 2 vertical; 6T: 4 vectored horizontal + 2 vertical	Conventional ROV frames

Voltage states of the battery are used as corners throughout: full (4.20 V/cell), nominal (3.70), low warning (3.50), cutoff (3.30) and absolute minimum (3.00). Unless stated, cases are evaluated at nominal 22.2 V; worst-case currents for constant-power loads are taken at cutoff, and worst-case thruster power at full charge.

3. Power architecture and tree

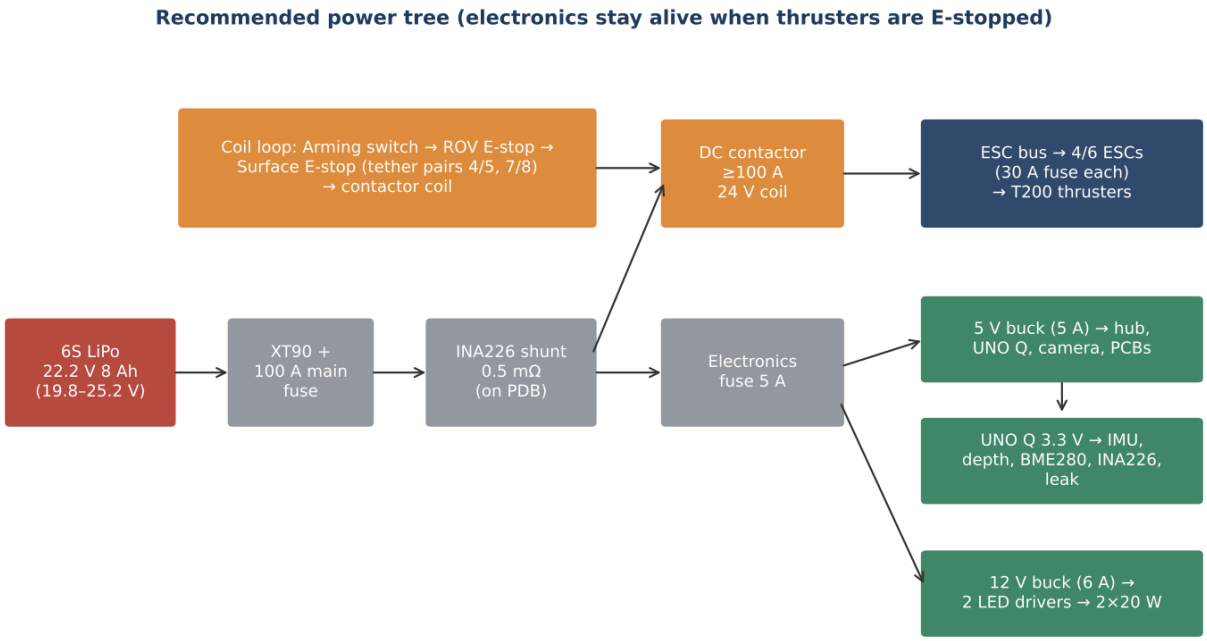


Figure 1 – Recommended power tree. Thrusters sit behind the contactor; electronics are fused separately upstream of it.

The list contains a single "Main Power/Arming Switch" and a DC contactor. The recommended interpretation is that the battery XT90 is the master disconnect (vehicle on), and the arming switch, the on-board E-stop and the surface E-stop are wired in series in the contactor coil loop. This has three electrical advantages: the switches carry only the coil current (≈131 mA) instead of up to 80 A; any stop command removes thruster power while the computer, camera, depth and leak sensing stay alive to report what happened; and a cut tether opens the loop, which is the correct fail-safe direction.

Node	Voltage range	Feeds	Max continuous current
Battery terminals	18.0–25.2 V (19.8 V operating floor)	Whole vehicle	80 A (4T at cap) / 98 A (6T budgeted)
ESC bus (post-contactor)	Battery minus I·R (see §10)	4 or 6 ESCs	Same as above minus hotel load
Electronics feed (pre-contactor, 5 A fuse)	Battery voltage	5 V and 12 V bucks, PDB indicators	2.85 A at cutoff, all loads peak
5 V rail	5.0 V ±3 %	Hub, UNO Q, camera, PCBs	2.25 A peak
3.3 V rail (UNO Q)	3.3 V	IMU, depth, BME280, INA226, leak	19.8 mA peak
12 V rail	12.0 V	2 LED drivers → 2 lamps	3.40 A at 100 % brightness
Coil loop	Battery voltage	Contactor coil via switches + tether	131 mA

4. Battery analysis (source)

Pack energy and capacity

$E_{nominal}$ = 8.0 Ah × 22.2 V = 177.6 Wh. Using 80 % depth of discharge, E_{usable} = 142.1 Wh and Q_{usable} = 6.4 Ah. At a 25C rating the pack can supply 200 A continuous; the highest design current (4T, all thrusters capped) is 80 A = 10.1C, so the pack is not the limiting element — the fuse, connector and ESCs are.

State	Per cell (V)	Pack (V)	Meaning / action
Full	4.20	25.2	Highest thruster stress; cap throttle to 72 %
Storage	3.80	22.8	Leave the pack here between sessions
Nominal	3.70	22.2	Reference point for all case tables
Low warning	3.50	21.0	Surface warning; begin return
Cutoff	3.30	19.8	Firmware disarms propulsion (under load, sag-compensated)
Absolute minimum	3.00	18.0	Damage threshold — never under load for > 1 s

Internal resistance and voltage sag

With 2.5 mΩ/cell the pack contributes 15 mΩ; the distribution path adds 3.84 mΩ, so R_{total} = 18.84 mΩ. Terminal voltage is $V = V_{oc} - I \cdot R$. For constant-power loads the current is the root of $R \cdot I^2 - V_{oc} \cdot I + P = 0$, which is how every case current in this report was solved.

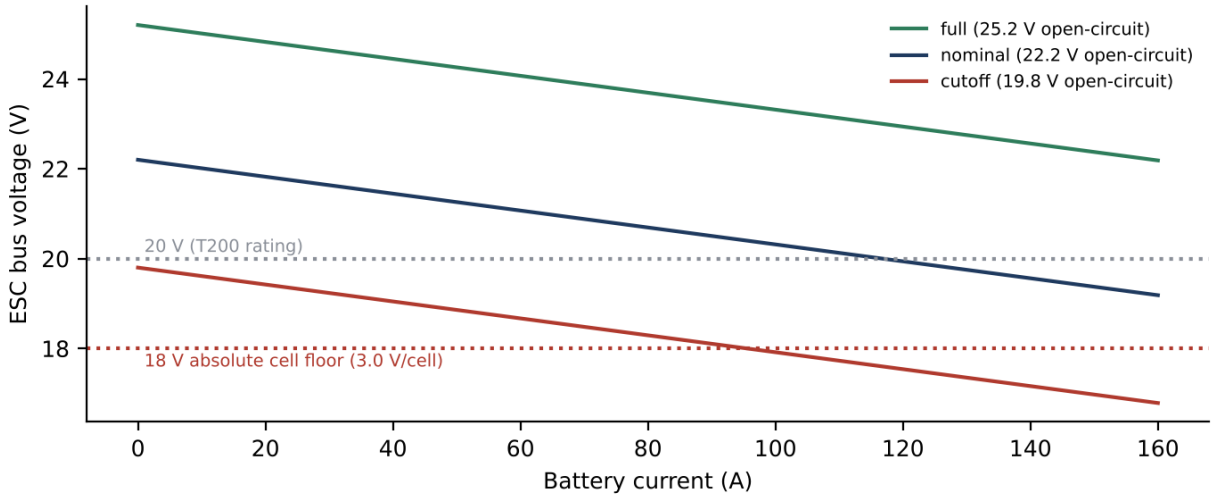


Figure 2 – ESC bus voltage versus battery current at three states of charge.

At cutoff with all four thrusters at cap the bus falls to 18.07 V while drawing 92.1 A — still above the 18 V cell-damage floor, but the low-voltage cutoff must use a sag-compensated estimate ($V_{oc} \approx V_{measured} + I \cdot R_{batt}$) or it will trip falsely during every hard manoeuvre and miss real depletion when idle.

Charging (Item 8, surface)

At 1C the charger delivers 8 A at up to 25.2 V, i.e. ≈202 W output (≈229 W from the mains at 88 % efficiency). A 0.5C charge (4 A, ≈100 W) roughly doubles the time but extends cycle life. Charge time from the 20 % DoD floor at 1C ≈ 0.8 h constant-current plus ≈0.3 h constant-voltage ≈ 1.1 h. Balance current on typical chargers is 0.3–1 A per cell — which is why a pack that comes home badly imbalanced should be balanced at storage voltage, not fast-charged. The charger must be rated ≥ 200 W for 1C on this pack; many "80 W" bench chargers will silently limit to ≈3 A.

5. Component-by-component analysis

Each item is listed with its supply, every operating state and the resulting voltage, current and power. Values are referred to the component's own supply rail; battery-side equivalents follow in Section 6.

5.1 Compute and communications (Items 1–4)

1. Arduino UNO Q (2 GB)

State	Supply V	Current	Power	Notes
Off	0 V	0	0	Board fully unpowered with vehicle off
Linux idle	5.0 V	≈0.3 A	≈1.5 W	QRB2210 idle, Wi-Fi disabled
Typical operation	5.0 V	0.70 A	3.5 W	Camera capture + encode, Ethernet stream, MCU control loop
Peak	5.0 V	1.50 A	7.5 W	All A53 cores loaded, GPU/encoder active, radio on
Rated maximum	5.0 V	3.0 A	15 W	USB-C PD contract (5 V/3 A only) — design ceiling

The board requests only a 5 V/3 A contract over USB-PD; VIN accepts 7–24 V but, when powered from VIN, the USB-C port does not act as a host — the hub, Ethernet adapter and camera would disappear. Feed it from the 5 V buck through the hub's power input (or the 5 V header pin) and verify host mode on the bench. Disabling Wi-Fi/Bluetooth (useless underwater) saves roughly 0.3–0.5 W and removes RF noise near the IMU.

2. USB-C hub with Gigabit Ethernet

State	Supply V	Current	Power	Notes
Link down	5.0 V	≈0.08 A	0.4 W	Hub controller + PHY in low-power
1000BASE-T active	5.0 V	0.18 A	0.9 W	Gigabit PHY dominates
100BASE-TX active	5.0 V	≈0.12 A	≈0.6 W	Recommended mode (see §12) — also saves ≈0.3 W
Peak	5.0 V	0.32 A	1.6 W	Link training + downstream USB enumeration

The hub is also the power path into the UNO Q. Its internal pass-through MOSFET and trace drop (typically 50–150 mΩ) matter: at 2 A total the drop is 0.1–0.3 V, which can push the board below its 4.75 V tolerance — set the 5 V buck to 5.1–5.2 V and measure at the UNO Q.

3. Shielded Cat6 Ethernet tether (50 m)

State	Supply V	Current	Power	Notes
Signal	±1 V differential	µA–mA	≪ 0.1 W	No power delivered over the tether in this design
DC resistance	—	—	—	23 AWG ≈ 66.6 mΩ/m → 3.33 Ω per conductor (50 m)
E-stop loop (if used)	12 V sense	≤ 20 mA	≤ 0.24 W	Pairs 4/5 + 7/8, loop ≈ 3.3 Ω, drop 67 mV at 20 mA

50 m is within the 100 m Ethernet channel limit. Ground the shield at the surface end only to avoid a ground loop through seawater and the ROV frame.

4. Waterproof Ethernet connector

State	Supply V	Current	Power	Notes
All states	—	≤ 20 mA (E-stop pins)	≈ 0	Passive; contact resistance ≈ 10–30 mΩ per pin is irrelevant at signal currents

Must expose all 8 conductors plus shield if the hardwired E-stop is routed through the tether. Electrical risk is moisture between pins — inspect and grease every dive.

5.2 Propulsion (Items 5–6)

5. Brushless DC thruster motors (×4 or ×6)

State	Supply V	Current	Power	Notes
Disarmed	0 V	0	0	Contactor open
Armed, neutral	Bus	0	0	ESC holds motor still
25 % throttle	Bus	0.3 A	6.2 W	Typical vertical trim
50 % throttle	Bus	2.3 A	50 W	Moderate manoeuvring
At cap (100 % command)	22.2 V	18.0 A	400 W	Firmware-limited
At cap, cutoff voltage	19.8 V	20.2 A	400 W	Constant power → current rises as voltage falls
Uncapped, 22.2 V	22.2 V	36.7 A	815 W	Exceeds 30 A ESC and 20 V rating
Uncapped, 25.2 V	25.2 V	43.0 A	1083 W	FAULT case C8
Stalled rotor	Bus	Up to ESC limit	—	Jammed by debris/tether; protect with a 30 A branch fuse and stall detection

Values are for the ESC+thruster pair measured at the ESC input (the basis of the published data). Throttle percentages are fractions of the capped command range. Full derivation is in Section 7.

6. Bidirectional ESCs (×4 or ×6)

State	Supply V	Current	Power	Notes
Disarmed	0 V	0	0	Behind contactor
Armed idle	22.2 V	50 mA	1.11 W	MCU + gate drivers; ×4 = 4.4 W, ×6 = 6.7 W
At cap	22.2 V	18.0 A	loss ≈ 16 W	≈4 % switching + conduction loss inside the hull
Close of contactor	25.2 V	inrush	—	Input capacitors (≈470 µF each assumed) → pre-charge required (§11)

Voltage rating must cover 25.2 V plus regenerative spikes during throttle reversal; a 26 V-rated ESC has almost no margin, so keep reversal ramps ≥ 150 ms in firmware and add a TVS / bulk capacitor on the ESC bus.

5.3 Energy source (Items 7–8)

7. 6S LiPo battery (22.2 V, 8000 mAh)

State	Supply V	Current	Power	Notes
Full	25.2 V	—	—	177.6 Wh nominal pack energy
Max design discharge	20.7 V	80.4 A	1663 W	4T at cap; internal loss 97 W
Mixed mission	≈22 V	12.3 / 18.0 A	263 / 375 W	4T / 6T average
Self-discharge, off	—	≈ 0	—	Unplug XT90 for storage; PDB LEDs alone would drain ≈ 0.07 Wh/day

See Section 4 for sag and voltage states.

8. 6S balance charger (surface)

State	Supply V	Current	Power	Notes
CC phase 1C	≤ 25.2 V	8.0 A	≈202 W out	≈230 W from mains
CC phase 0.5C	≤ 25.2 V	4.0 A	≈101 W	Recommended routine charge
Balance	4.2 V/cell	0.3–1 A	≤ 4 W	Per-cell bleed

Surface equipment; not part of the vehicle budget. Charge only in a LiPo bag and never inside the sealed hull.

5.4 Sensors and camera (Items 9–14)

9. BNO055 IMU

State	Supply V	Current	Power	Notes
NDOF fusion	3.3 V	12.3 mA	40.6 mW	9-axis fusion at 100 Hz
Low-power mode	3.3 V	≈0.3 mA	≈1 mW	Not used in operation
Suspend	3.3 V	40 µA	0.13 mW	Between dives

Keep ≥ 10 cm from ESC/thruster leads; motor current produces magnetic fields that corrupt the magnetometer — this is the most common "electrical" cause of heading drift.

10. MS5837 depth/pressure

State	Supply V	Current	Power	Notes
Conversion	3.3 V	1.25 mA	4.1 mW	≈9 ms per D1/D2 at OSR 4096
20 Hz average	3.3 V	≈0.9 mA	≈3 mW	Duty-cycled
Standby	3.3 V	≈0.1 µA	≈0	—

Negligible power. Put it on its own I²C segment or keep the bus short; it shares the I²C address space with nothing else here.

11. Leak detection probes (×3)

State	Supply V	Current	Power	Notes
Dry	3.3 V	≈0	≈0	Open circuit
Wet (each)	3.3 V	0.33 mA	1.1 mW	10 kΩ pull-up into water path
All wet	3.3 V	1.0 mA	3.3 mW	Triggers emergency ascent C7

Use AC or pulsed excitation if possible — continuous DC through wet probes electro-plates the contacts and they lose sensitivity over a season.

12. BME280 temperature & humidity (hull)

State	Supply V	Current	Power	Notes
Forced 1 Hz	3.3 V	3.6 µA avg	12 µW	T + P + H
Measuring	3.3 V	0.71 mA peak	2.4 mW	During pressure conversion
Sleep	3.3 V	0.1 µA	≈0	—

Rising humidity inside the hull is an early leak indicator; log it at 1 Hz.

13. INA226 voltage/current monitor

State	Supply V	Current	Power	Notes
Continuous	3.3 V	0.33 mA	1.1 mW	IC supply
Shunt at 80 A	—	80 A	3.2 W	0.5 mΩ shunt dissipation — use a ≥ 5 W part
Shunt at 20 A	—	20 A	0.20 W	Typical cruise

Bus-voltage input tolerates 36 V, covering 25.2 V. Configuration is derived in Section 13.

14. 1080p low-light USB camera

State	Supply V	Current	Power	Notes
Streaming	5.0 V	0.20 A	1.0 W	1080p30
Peak	5.0 V	0.30 A	1.5 W	Enumeration / auto-exposure
Suspended	5.0 V	≈2.5 mA	≈12 mW	USB suspend

At 1080p30, MJPEG output can exceed the 100 Mbit/s link proposed in §12; use H.264 or 720p if the stream stutters.

5.5 Lighting (Items 15–16)

15. Waterproof LED lights (12 V, 20 W) ×2

State	Supply V	Current	Power	Notes
Off	12 V	0	0	
50 % dim	12 V	0.83 A each	10 W each	PWM duty 50 %
100 %	12 V	1.67 A each	20 W each	40 W total

LED lamps rely on water for cooling. Running them at full power in air for more than a minute or two can overheat them — firmware should limit to ≤ 20 % out of water (use depth < 0.2 m as the trigger).

16. LED drivers ×2

State	Supply V	Current	Power	Notes
Off	12 V	≈1 mA	≈0.01 W	Standby
100 %	12 V	1.70 A	loss ≈ 0.4 W	Per driver; ≈98 % efficient PWM stage

If the drivers turn out to be constant-current bucks instead of PWM switches, use 90 % efficiency: 12 V rail load rises to ≈44 W and the 12 V buck sizing in §6 still holds.

5.6 Power conversion (Items 17–18)

17. 5 V buck converter

State	Supply V	Current	Power	Notes
Typical	in 22.2 V / out 5 V	in 0.28 A / out 1.13 A	loss 0.63 W	
Peak, at cutoff	in 19.8 V / out 5 V	in 0.63 A / out 2.25 A	loss 1.25 W	
Rating required	in ≥ 30 V	out ≥ 5 A	≥ 25 W	≈ 2× margin for USB inrush and UNO Q peaks

Input must survive 25.2 V plus ESC-bus transients; a 36 V-rated input part is preferred even though the electronics feed is upstream of the contactor.

18. 12 V buck converter

State	Supply V	Current	Power	Notes
Lights off	in 22.2 V	≈10 mA	≈0.2 W	Quiescent
Lights 100 %	in 22.2 V / out 12 V	in 1.98 A / out 3.40 A	loss 3.1 W	
Lights 100 %, cutoff	in 19.8 V	in 2.22 A	43.9 W in	
Rating required	in ≥ 30 V	out ≥ 6 A	≥ 72 W	≈ 75 % load at full brightness

This converter is the largest heat source among the electronics (≈3 W at full brightness); mount it on the hull end-cap or a metal plate.

5.7 Protection and switching (Items 19–22)

19. Main fuse and fuse holders (set)

State	Supply V	Current	Power	Notes
Main 100 A	Bus	≤ 80 A design	4.5 W at 80 A	≈0.7 mΩ with holder
ESC branch 30 A ×4/6	Bus	≤ 20.2 A	≈0.3 W each	Isolates a shorted ESC
Electronics 5 A	Bus	≤ 2.85 A	≈0.02 W	

Fuses must have a DC rating ≥ 32 V. Coordination is analysed in Section 11.

20. DC contactor

State	Supply V	Current	Power	Notes
Open	—	0	0	Thrusters dead
Coil energised, full	25.2 V	131 mA	3.31 W	Coil heats inside hull continuously
Coil energised, cutoff	19.8 V	103 mA	2.04 W	Must stay ≥ hold voltage
Main contacts at 80 A	—	80 A	3.2 W	≈0.5 mΩ

Choose a contactor with a DC breaking rating at ≥ 100 A / 32 V: an E-stop during full thrust interrupts a large inductive DC current. A coil economiser (PWM hold) cuts coil heat by ≈70 %.

21. Emergency stop switch (on vehicle)

State	Supply V	Current	Power	Notes
Closed (run)	Coil loop	131 mA	≈0	NC contact, DC rated ≥ 1 A / 30 V
Pressed	—	0	0	Coil drops → contactor opens in ≈10–20 ms

Because it only carries coil current, a small marine-rated NC mushroom switch is sufficient.

22. Main power/arming switch

State	Supply V	Current	Power	Notes
Disarmed	—	0	0	
Armed	Coil loop	131 mA	≈0	Series with both E-stops

If instead this switch is wired in the main battery line it must carry 80 A and break DC under load — a much harder (and larger) part. The coil-loop architecture is strongly preferred.

5.8 PCBs (Items 23–25)

23. Power distribution PCB

State	Supply V	Current	Power	Notes
Main path	Bus	≤ 80 A	5.1 W (copper + shunt)	2 oz copper needs ≈42 mm width for 80 A at 20 °C rise (IPC-2221)
ESC branch	Bus	≤ 20 A	—	≈6 mm of 2 oz copper per branch
Indicators	Bus	3 mA	0.07 W	

For the main path, use bolted copper bus bars or both outer layers in parallel with stitching vias; a single 41 mm pour is impractical on a compact board.

24. Interface PCB

State	Supply V	Current	Power	Notes
Operating	5 V / 3.3 V	≈30 mA	0.15 W	Level shifting, PWM fan-out to ESCs, status LEDs
Peak	5 V	60 mA	0.30 W	All LEDs on

Route ESC PWM signals with a ground return alongside each; keep it away from the IMU.

25. Leak detection PCB

State	Supply V	Current	Power	Notes
Dry	3.3–5 V	≈10 mA	0.05 W	Comparators/Schmitt inputs
Leak alarm	5 V	≈50 mA	0.25 W	LED + buzzer

Hardware OR of the three probes to a dedicated MCU interrupt pin, so a leak is detected even if Linux has crashed.

5.9 Surface equipment and connectors (Items 26–28)

26. USB joystick/gamepad (surface)

State	Supply V	Current	Power	Notes
Active	5 V (laptop USB)	≈50–100 mA	≤ 0.5 W	Not on vehicle budget

Electrically trivial; the relevant risk is the laptop going to sleep — disable USB selective suspend.

27. Surface emergency stop box

State	Supply V	Current	Power	Notes
Run	12 V sense from ROV	≤ 20 mA	≤ 0.24 W	Loop through tether pairs
Stop	—	0	0	Loop opens → ROV contactor drops

Fail-safe: tether break, connector unplug or a pressed button all produce the same safe state.

28. Power connectors (XT60/XT90)

State	Supply V	Current	Power	Notes
XT90 at 80 A	Bus	80 A	3.8 W	Battery main; rated ≈90 A continuous
XT60 at 20 A	Bus	20 A	0.32 W	ESC branches/electronics; rated ≈60 A
Plug-in spark	25.2 V	inrush	—	XT90-S (anti-spark) recommended for the battery

Connector heating scales with I^2: a worn XT90 at 2 mΩ would dissipate 13 W at 80 A. Check for discoloured housings after hard runs.

6. Rail budgets and converter sizing

3.3 V sensor rail

Load	Typical	Peak
BNO055 IMU (NDOF fusion)	40.59 mW	49.50 mW
MS5837 depth (20 Hz, OSR 4096)	2.97 mW	4.12 mW
BME280 (1 Hz forced)	0.01 mW	2.36 mW
INA226 monitor	1.09 mW	1.39 mW
I2C pull-ups 4.7 k x2 (50% low)	2.31 mW	4.62 mW
Leak probes x3 (10 k pull-up, wet)	0.00 mW	3.27 mW
Total	**47.0 mW (14.2 mA)**	**65.3 mW (19.8 mA)**

The rail is supplied by the UNO Q's own 3.3 V output. Even assuming a linear regulator (66 % efficient from 5 V), the 5 V cost is only 71 mW typical.

5 V rail

Load	Typical (W)	Peak (W)	Typical (A)
Arduino UNO Q (2 GB) board	3.50	7.50	0.70
USB-C hub + GbE PHY	0.90	1.60	0.18
1080p low-light USB camera	1.00	1.50	0.20
3.3 V sensor rail (via UNO Q, linear worst case)	0.07	0.10	0.01
Interface PCB (buffers, LEDs)	0.15	0.30	0.03
Leak detection PCB (comparators, LED/buzzer)	0.05	0.25	0.01
Total output	**5.67**	**11.25**	**1.13**
Buck input at 22.2 V (η = 90 %)	6.30 W / 0.284 A	12.50 W / 0.563 A	
Buck input at 19.8 V (worst current)	0.318 A	0.631 A	

12 V rail

Brightness	Rail output	Buck input @22.2 V	Buck input @19.8 V	Buck loss
0 %	0.0 W / 0.00 A	0.00 A	0.00 A	0.00 W
25 %	10.2 W / 0.85 A	0.49 A	0.55 A	0.77 W
50 %	20.4 W / 1.70 A	0.99 A	1.11 A	1.54 W
75 %	30.6 W / 2.55 A	1.48 A	1.66 A	2.30 W
100 %	40.8 W / 3.40 A	1.98 A	2.22 A	3.07 W

Battery bus loads (direct)

Load	22.2 V	25.2 V	19.8 V
ESC idle ×4	4.44 W	5.04 W	3.96 W
ESC idle ×6	6.66 W	7.56 W	5.94 W
Contactor coil	2.57 W	3.31 W	2.04 W
PDB indicators	0.067 W	0.076 W	0.059 W

Observation

Arming the vehicle more than doubles the idle draw (C1 → C2: 6.4 W → 13.4 W for 4T) because ESC quiescent current and the contactor coil together exceed the entire computing stack. Disarm whenever the ROV is parked on the surface between tasks.

7. Propulsion deep-dive: thrusters on 6S

Published full-throttle points for the T200 are 205 W at 12 V, 390 W at 16 V and 645 W at 20 V, with 20 V as the maximum recommended voltage. Fitting $P = a \cdot V^k$ through these points gives k = 2.24. Power rises faster than V^2 because the motor spins faster and the propeller load grows with roughly the cube of RPM.

$$P_{full}(V) = 645\ W \times (V / 20\ V)^{2.24}$$

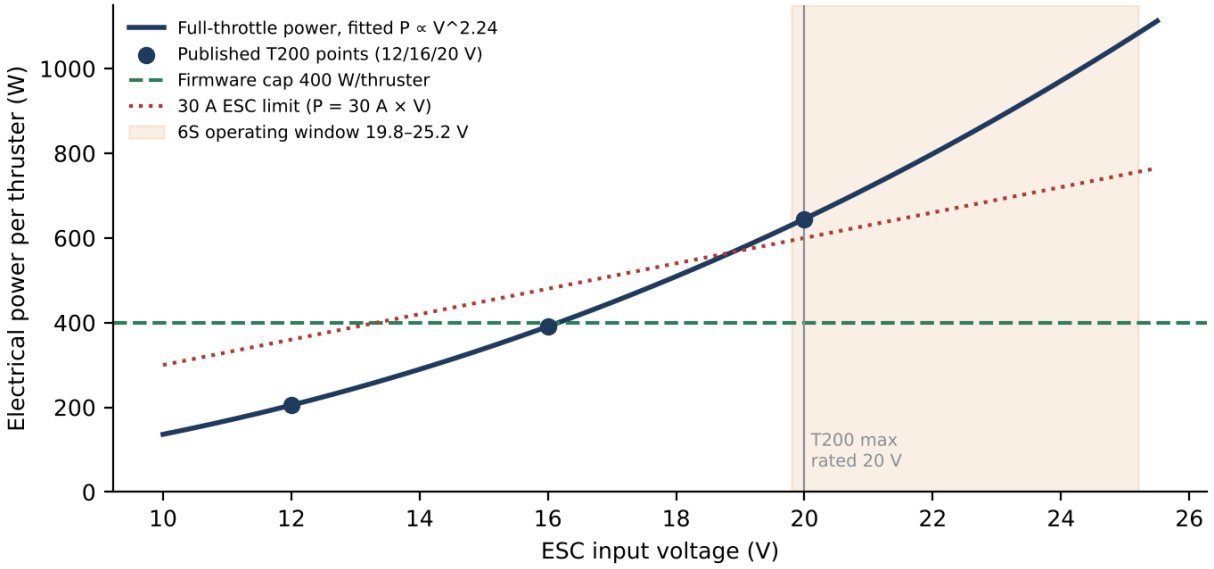


Figure 3 – Thruster power versus voltage, with the 6S window, ESC current limit and firmware cap.

Bus voltage	Full-throttle P	Full-throttle I	Max throttle for 400 W cap	Current at cap	Thrust at cap (est.)
19.8 V	631 W	31.8 A	86 %	20.2 A	≈5.34 kgf
21.0 V	720 W	34.3 A	82 %	19.0 A	≈5.34 kgf
22.2 V	815 W	36.7 A	79 %	18.0 A	≈5.34 kgf
23.0 V	883 W	38.4 A	77 %	17.4 A	≈5.34 kgf
24.0 V	971 W	40.5 A	74 %	16.7 A	≈5.34 kgf
25.2 V	1083 W	43.0 A	72 %	15.9 A	≈5.34 kgf

Why a power cap rather than a fixed throttle cap: a fixed 72 % throttle limit is safe at 25.2 V but wastes thrust as the battery drains. Capping power (by scaling the maximum command with measured bus voltage) keeps the thruster at ≈400 W — the equivalent of a T200 at its 16 V nominal point, ≈5.3 kgf forward thrust — over the entire discharge. The trade-off is that current rises as voltage falls, peaking at 20.2 A per thruster at cutoff, which is still inside the 30 A ESC rating with margin.

4 vs 6 thrusters

Metric	4-thruster	6-thruster
Total installed capped power	1600 W	2400 W
Battery current, all at cap (22.2 V)	80.4 A	124.1 A
Battery current, all at cap (19.8 V)	92.1 A	n/a — exceeds fuse; global budget applies
With 1700 W global budget (6T)	not needed	85.8 A @22.2 V, 98.4 A @19.8 V
ESC idle (armed)	4.4 W	6.7 W
Cruise C5 current	11.4 A	19.5 A
Controllable DOF	Surge, yaw, heave (+roll/pitch if verticals split)	Surge, sway, yaw, heave

Six thrusters add sway control and redundancy, but the 6T vehicle can only use its extra installed power if the electrical path is upgraded (AS150 connector, 150 A fuse, 6 AWG). Within the listed parts (XT90, one main fuse), the 6T build must run with a 1700 W global propulsion budget, distributed by the thrust allocator.

8. Operating-case study (C1–C8)

Each case defines a throttle for horizontal (u_H) and vertical (u_V) thrusters as a fraction of the capped command, plus a light level. Thruster power per unit is $P_{cap} \cdot u^3$. All electronics, converter losses, ESC idle and the coil are included. Battery current is solved including I²R sag.

Case	Description	u_H	u_V	Lights	Armed
C1	Disarmed on deck / pre-dive checks	0 %	0 %	0 %	No
C2	Armed, thrusters neutral (idle in water)	0 %	0 %	0 %	Yes
C3	Station-keeping, lights 50 %	15 %	30 %	50 %	Yes
C4	Inspection: hold + lights 100 %	20 %	35 %	100 %	Yes
C5	Cruise: horizontals 60 %, verticals 30 %	60 %	30 %	100 %	Yes
C6	Aggressive manoeuvre: all at cap	100 %	100 %	100 %	Yes
C7	Emergency ascent: verticals at cap	0 %	100 %	100 %	Yes
C8	FAULT: cap disabled, full throttle	100 %	100 %	100 %	Yes

Case results – 4-thruster configuration

Case	Thrusters	ESC idle	Coil	5 V buck in	12 V buck in	Total	Battery I	Bus V	I²R loss
C1	0	0.0	0.0	6.3	0.0	**6 W**	**0.3 A**	22.19 V	0.0 W
C2	0	4.4	2.6	6.3	0.0	**13 W**	**0.6 A**	22.19 V	0.0 W
C3	24	4.4	2.6	6.3	21.9	**60 W**	**2.7 A**	22.15 V	0.1 W
C4	41	4.4	2.6	6.3	43.9	**98 W**	**4.4 A**	22.12 V	0.4 W
C5	194	4.4	2.6	6.3	43.9	**252 W**	**11.4 A**	21.98 V	2.5 W
C6	1600	4.4	2.6	12.5	43.9	**1663 W**	**80.4 A**	20.68 V	121.8 W
C7	800	4.4	2.6	12.5	43.9	**863 W**	**40.3 A**	21.44 V	30.6 W
C8	4334	5.0	3.3	12.5	43.9	**4398 W**	**206.4 A**	21.31 V	802.4 W

Case results – 6-thruster configuration

Case	Thrusters	ESC idle	Coil	5 V buck in	12 V buck in	Total	Battery I	Bus V	I²R loss
C1	0	0.0	0.0	6.3	0.0	**6 W**	**0.3 A**	22.19 V	0.0 W
C2	0	6.7	2.6	6.3	0.0	**16 W**	**0.7 A**	22.19 V	0.0 W
C3	27	6.7	2.6	6.3	21.9	**65 W**	**2.9 A**	22.15 V	0.2 W
C4	47	6.7	2.6	6.3	43.9	**107 W**	**4.8 A**	22.11 V	0.4 W
C5	367	6.7	2.6	6.3	43.9	**427 W**	**19.5 A**	21.83 V	7.2 W
C6	2400	6.7	2.6	12.5	43.9	**2466 W**	**124.1 A**	19.86 V	290.4 W
C7	800	6.7	2.6	12.5	43.9	**866 W**	**40.4 A**	21.44 V	30.7 W
C8	6500	7.6	3.3	12.5	43.9	**6568 W**	**354.7 A**	18.52 V	2369.7 W

Power in W unless shown. C1–C7 at 22.2 V; C8 at full charge 25.2 V. Highlighted rows exceed the 100 A main fuse.

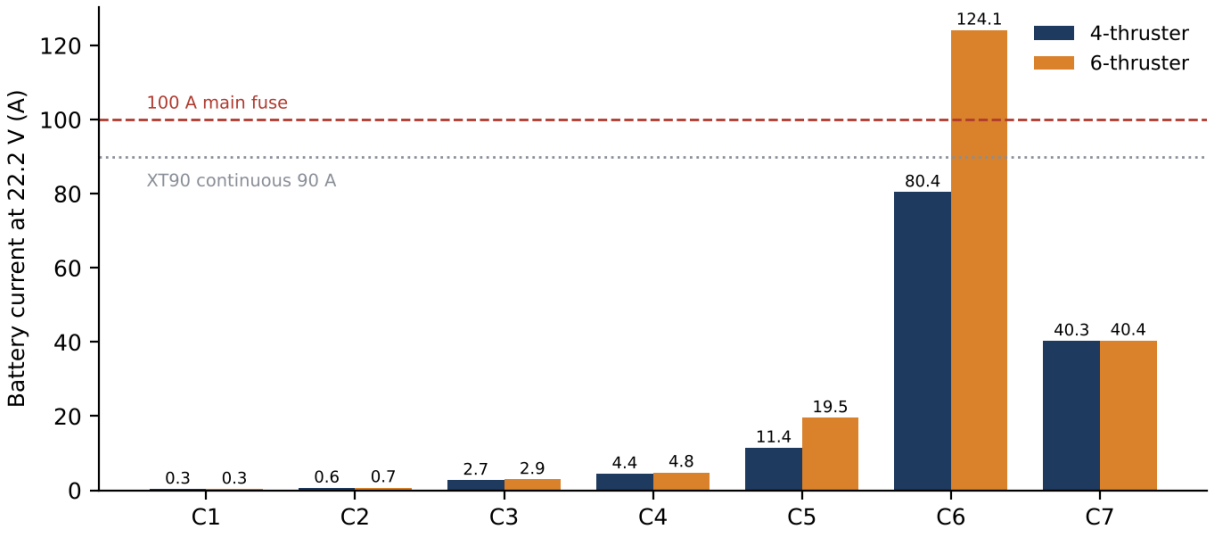


Figure 4 – Battery current per case (C8 omitted: 206 A / 355 A).

Reading the cases

- **C1–C2:** sub-1 A. The vehicle could idle armed for 10.6 h on usable capacity — endurance is never limited by electronics.
- **C3–C4:** lights dominate. At 100 % lights, the two lamps are 45 % of the C4 (4T) budget. Dimming to 50 % is the single most effective endurance lever during inspection.
- **C5:** cubic throttle law makes cruise cheap — 60 % throttle uses only 22 % of capped power per thruster.
- **C6:** 4T is safely below the 100 A fuse (80 A); 6T is not (124 A) without the global budget.
- **C7:** emergency ascent uses only the verticals, ≈40 A, well inside every rating — so a leak alarm can always command full ascent.
- **C8 (fault):** if the cap is bypassed (bug, wrong PWM range, ESC recalibration) at full charge, demand is 206 A (4T) — the main fuse clears in seconds, ESCs are over-current and thrusters see 126 % of rated voltage. This is the failure the firmware cap and branch fuses exist to prevent.

Voltage-corner sensitivity (4T, C6)

Open-circuit V	Battery I	Bus V under load	I²R loss	Hotel share
25.2 V	69.7 A	23.89 V	92 W	3.4 %
22.2 V	80.4 A	20.68 V	122 W	3.4 %
21.0 V	85.8 A	19.38 V	139 W	3.4 %
19.8 V	92.0 A	18.07 V	160 W	3.4 %

9. Endurance and mission profile

Runtime per case = Q_{usable} / $I_{battery}$, with Q_{usable} = 6.4 Ah. A realistic dive mixes cases; the weighted profile below represents a typical inspection mission.

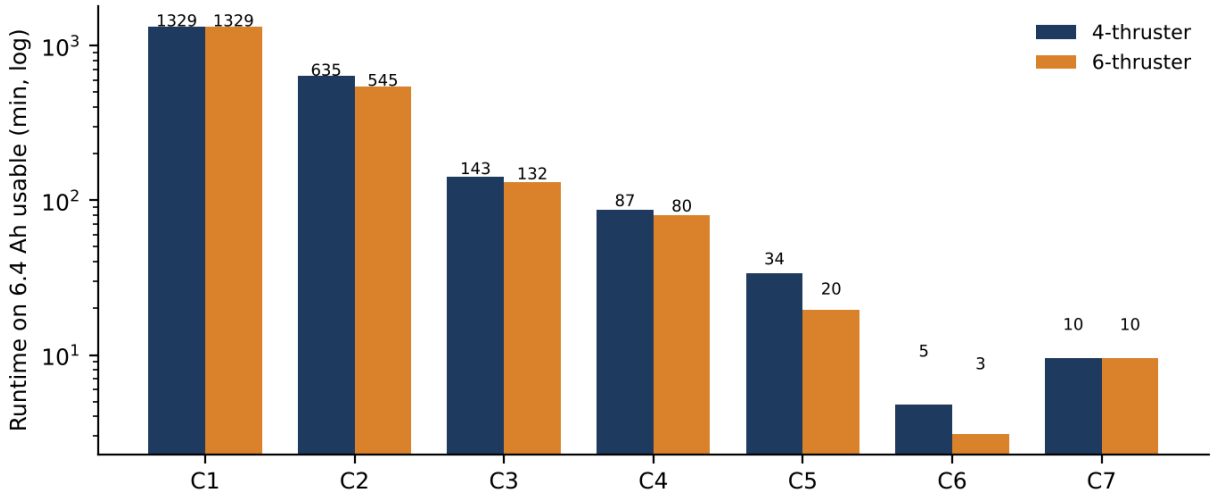


Figure 5 – Continuous runtime per case (log scale).

Case	Weight	4T current	4T contribution	6T current	6T contribution
C2	5 %	0.6 A	0.03 A	0.7 A	0.04 A
C3	25 %	2.7 A	0.67 A	2.9 A	0.73 A
C4	30 %	4.4 A	1.33 A	4.8 A	1.45 A
C5	30 %	11.4 A	3.43 A	19.5 A	5.86 A
C6	7 %	80.4 A	5.63 A	124.1 A	8.69 A
C7	3 %	40.3 A	1.21 A	40.4 A	1.21 A
Weighted	100 %		**12.30 A**		**17.97 A**
Endurance			**31 min**		**21 min**

The 7 % of time spent at full manoeuvre consumes roughly half the energy of the mission: for 4T, C6 contributes 5.6 A of the 12.3 A average. Pilot behaviour — not electronics — sets endurance. Recommended practice: a 15 % energy reserve for recovery, giving a planning dive time of ≈27 min (4T) / ≈18 min (6T).

State-of-charge estimation

Voltage alone is a poor SoC indicator under load because of sag (up to 1.5 V in C6). The INA226 enables coulomb counting: $SoC = SoC_0 - \int I\,dt / 8\ Ah$, re-anchored to the open-circuit voltage after ≥ 2 minutes of rest at disarm. With a 5 mA current LSB the integration error is negligible compared with the ±5 % capacity tolerance of the pack.

10. Losses, voltage drop and thermal load

Distribution path resistance

Element	Resistance	Loss at 20 A	Loss at 50 A	Loss at 80 A
XT90 battery connector (pair)	0.60 mΩ	0.24 W	1.50 W	3.84 W
Main fuse 100 A + holder	0.70 mΩ	0.28 W	1.75 W	4.48 W
DC contactor main contacts	0.50 mΩ	0.20 W	1.25 W	3.20 W
INA226 shunt 0.5 mOhm	0.50 mΩ	0.20 W	1.25 W	3.20 W
8 AWG main leads, 0.6 m loop (2.06 mOhm/m)	1.24 mΩ	0.50 W	3.10 W	7.94 W
PDB copper (2 oz, est.)	0.30 mΩ	0.12 W	0.75 W	1.92 W
Distribution total	**3.84 mΩ**	1.54 W	9.60 W	**24.6 W**
Battery internal	15.0 mΩ	6.0 W	37.5 W	96 W

Every milliohm in the main path costs 6.4 W at 80 A. The shunt and the fuse are the two largest discretionary contributors; a 0.25 mΩ shunt halves the shunt loss at the cost of half the INA226 resolution.

Heat released inside the hull

Source	C4 (4T)	C5 (4T)	C6 (4T)	C6 (6T, budgeted)
ESCs (switching loss + idle)	6.1 W	12.2 W	68.4 W	74.7 W
5 V electronics (load + buck loss)	6.3 W	6.3 W	12.5 W	12.5 W
12 V buck loss	3.1 W	3.1 W	3.1 W	3.1 W
Contactor coil	2.6 W	2.6 W	2.6 W	2.6 W
Distribution I^2R (in-hull)	0.1 W	0.5 W	24.8 W	28.3 W
Total in hull	**18.1 W**	**24.7 W**	**111.4 W**	**121.1 W**

Steady-state hull heat is ≈10–25 W in normal operation and short bursts of 70–90 W in full manoeuvre. Water-cooled aluminium end-caps handle this easily in water; the critical case is **on deck in sunlight with electronics running**, where the only cooling is air. Bolt the ESCs and 12 V buck to the end-cap and keep deck time with the lights off.

11. Protection coordination: fuses, wiring, contactor, pre-charge

Branch	Design current	Rule	Selected	Wire
Battery main	80 A (4T C6) / 98 A (6T budget, cutoff)	1.25 × continuous, slow-blow	100 A MIDI/ANL, ≥ 32 V DC	8 AWG silicone (≈ 2.1 mΩ/m)
Each ESC	20.2 A (cap at 19.8 V)	1.25 × 20.2 = 25.3 A	30 A ATO blade	14 AWG
Electronics feed	2.85 A peak	≈1.5–2 ×	5 A blade	20 AWG
5 V output	2.25 A peak	Polyfuse	3 A PTC to hub	20 AWG
12 V lights	1.70 A each	≈2 ×	3 A blade each	20 AWG
Coil loop	131 mA	≈4 ×	0.5 A fuse	24 AWG / tether pairs

Selectivity: a shorted ESC branch draws hundreds of amps for milliseconds; a 30 A blade clears in < 10 ms at 10× rating, while a 100 A slow-blow needs seconds at 2×, so the branch fuse opens first and the vehicle keeps its remaining thrusters and all electronics.

Contactor rating

- Continuous ≥ 100 A, DC-rated contacts (magnetic blow-out), ≥ 32 V DC breaking capacity at ≥ 100 A.
- Coil 24 V nominal with pick-up ≤ 18 V so it pulls in at cutoff (19.8 V) and holds through sag; coil current 103–131 mA across the window.
- Flyback diode + Zener (or TVS) across the coil: a bare diode slows release to ≈50 ms; diode + 24 V Zener restores ≈10–15 ms E-stop response.

Pre-charge

Total ESC-bus capacitance (≈470 µF per ESC ×6 + 2 × 100 µF) ≈ 3.02 mF. Closing the contactor directly gives a theoretical peak of V/R = 25.2 V / 18.8 mΩ ≈ 1338 A, which pits and welds contactor contacts over time and is the usual cause of "contactor stuck closed" — a serious E-stop failure.

$$\tau = R_{pre} \cdot C = 10\ \Omega \times 3.02\ \text{mF} = 30.2\ \text{ms};\ 5\tau = 151\ \text{ms}$$
$$\text{Peak pre-charge current} = 25.2\ \text{V} / 10\ \Omega = 2.5\ \text{A};\ \text{energy in resistor} = \tfrac{1}{2}CV^2 = 0.96\ \text{J}$$

A 10 Ω, 5 W resistor in series with a small pre-charge relay, wired across the contactor contacts and switched off once the contactor closes, with a 200 ms firmware delay before ESC commands are accepted is sufficient. If space is tight, an XT90-S anti-spark connector protects the battery connector but does not protect the contactor.

Wire voltage drop check

Run	Gauge	Length (loop)	Current	Drop	% of 22.2 V
Battery → PDB	8 AWG	0.6 m	80 A	0.10 V	0.4 %
PDB → ESC	14 AWG	0.4 m	20.2 A	0.07 V	0.3 %
ESC → thruster (3-phase)	16 AWG	1.0 m	≈20 A rms	≈0.26 V	—
12 V → lamp	20 AWG	1.2 m	1.7 A	0.07 V	0.6 % of 12 V
5 V → hub	20 AWG	0.3 m	2.3 A	0.023 V	0.5 % of 5 V

12. Tether, Ethernet and emergency-stop loop

1000BASE-T transmits on all four pairs simultaneously, so a Gigabit link leaves no spare conductors in the Cat6 tether. 100BASE-TX uses only pairs 1/2 and 3/6, freeing 4/5 and 7/8. A compressed 1080p video stream needs on the order of 4–10 Mbit/s; telemetry and control are a few hundred kbit/s, so 100 Mbit/s is ample when the camera stream is H.264 (or MJPEG at reduced resolution). Force 100 Mbit/s full-duplex on the surface NIC or hub with ethtool.

```
Per conductor (23 AWG): 0.0666 Ω/m × 50 m = 3.33 Ω
Pair as one leg (two in parallel): 1.67 Ω; loop (go + return) = 3.33 Ω
Sense current 20 mA → drop = 67 mV; loop power = 1.33 mW
```

Recommended circuit: the ROV sources 12 V through a 470 Ω resistor into the loop (pair 4/5), through the normally-closed surface E-stop, back on pair 7/8 into an optocoupler (≈20 mA). The optocoupler output and the local switches drive a small relay/MOSFET in series with the contactor coil. The contactor coil (131 mA) is never driven over the tether directly — at that current the loop would drop 0.44 V and, more importantly, a long inductive loop would pick up thruster noise.

Fail-safe behaviour

Tether severed, connector flooded open, surface box unplugged or button pressed → loop current stops → optocoupler off → contactor opens → thrusters unpowered, electronics still alive. In addition, the firmware heartbeat (§13) neutralises the thrusters within 500 ms of losing the Ethernet control stream, covering the case where the hardware loop is intact but the software link is not.

13. Firmware power management (embedded implementation)

The UNO Q splits naturally: the STM32U585 (Zephyr) owns hard real-time power safety; Linux on the QRB2210 owns video, networking and the pilot interface. Power protection must never depend on Linux being alive.

INA226 configuration

```
Shunt 0.5 mΩ; full-scale shunt voltage 81.92 mV → I_max = 163.8 A
Current_LSB = 5 mA (≥ I_max / 2^15 = 5.00 mA)
CAL = 0.00512 / (Current_LSB × R_shunt) = 0.00512 / (0.005 × 0.0005) = 2048 (0x0800)
Power_LSB = 25 × Current_LSB = 0.125 W; Bus-voltage LSB = 1.25 mV
Conversion 1.1 ms (V_shunt) + 1.1 ms (V_bus), AVG = 16 → update every ≈35 ms (≈28 Hz)
```

Power-limiter loop (50 Hz on STM32U585)

```
/* power_mgr.c - runs at 50 Hz on the STM32U585 (Zephyr thread, prio high) */
#define P_THR_CAP_W     400.0f   /* per-thruster electrical cap            */
#define P_BUS_BUDGET_W 1700.0f   /* global propulsion budget (6T build)    */
#define V_CUTOFF_OC     19.8f    /* 3.30 V/cell, open-circuit estimate     */
#define V_WARN_OC       21.0f    /* 3.50 V/cell                            */
#define R_BATT_OHM      0.015f   /* measured pack IR - update from test    */
#define HEARTBEAT_MS    500U

static volatile float g_vbus, g_ibus;       /* written by INA226 alert ISR */

void power_mgr_step(float cmd[], int n)     /* cmd: -1..+1 per thruster    */
{
    float v = g_vbus, i = g_ibus;
    float v_oc = v + i * R_BATT_OHM;           /* sag-compensated SoC voltage */

    /* 1. per-thruster cap: P_full(V) = 645*(V/20)^2.24, P ~ u^3          */
    float p_full = 645.0f * powf(v / 20.0f, 2.24f);
    float u_max  = fminf(1.0f, cbrtf(P_THR_CAP_W / p_full));

    /* 2. global budget: scale all commands together (keeps thrust vector)  */
    float p_req = 0.0f;
    for (int k = 0; k < n; k++) p_req += P_THR_CAP_W * fabsf(cmd[k]*cmd[k]*cmd[k]);
    float s = (p_req > P_BUS_BUDGET_W) ? cbrtf(P_BUS_BUDGET_W / p_req) : 1.0f;

    /* 3. low-voltage derate and cutoff                                    */
    if (v_oc < V_WARN_OC)  s *= 0.6f;
    if (v_oc < V_CUTOFF_OC || heartbeat_age_ms() > HEARTBEAT_MS) s = 0.0f;

    for (int k = 0; k < n; k++) esc_write(k, cmd[k] * s * u_max);
    iwdg_refresh();                          /* watchdog: 100 ms timeout    */
}
```

Additional firmware rules derived from this analysis

- Slew-limit each ESC command to ≤ 100 % per 150 ms, and pass through zero before reversing — limits regenerative bus spikes toward the 26 V ESC rating.
- Accept ESC commands only ≥ 200 ms after the contactor closes (pre-charge complete) and only with bus voltage ≥ 19.8 V.
- Leak interrupt (any probe) → latch C7 emergency ascent: verticals at cap, horizontals 0, lights 100 %, alarm to surface.
- Lights limited to 20 % while depth < 0.2 m (lamps are water-cooled).
- Log V, I, P, SoC, hull temperature and humidity at 10 Hz to eMMC for post-dive review — this is the data that will validate (or correct) this report.

Resource usage estimate (MCU side)

Resource	Estimate	Notes
Flash	≈ 18 kB	INA226, BNO055, MS5837, BME280 drivers + power manager + ESC PWM
RAM	≈ 6 kB	Thread stacks 2 × 1.5 kB, sensor buffers, 1 s log ring
CPU	< 3 % at 160 MHz	50 Hz loop with two powf/cbrtf per cycle on the FPU
Timers	1 × 4-ch + 1 × 2-ch	ESC PWM at 50–400 Hz
Interrupts	INA226 ALERT, leak OR, E-stop status	All short ISRs that set flags; work deferred to threads

14. Post-mortem findings register

ID	Severity	Finding	Impact	Recommendation
F1	Critical	6S full charge (25.2 V) exceeds 20 V thruster rating; uncapped draw 1083 W / 43 A per thruster	ESC over-current, motor overheating, fuse trips	Voltage-compensated 400 W/thruster cap in MCU firmware; or select 24 V-rated thrusters
F2	High	6 thrusters at cap demand 124 A	Exceeds 100 A fuse and XT90	Global 1700 W budget, or upgrade to AS150 + 150 A fuse + 6 AWG
F3	High	UNO Q powered via VIN disables USB host	No camera, no Ethernet hub	Power through hub/PD input or 5 V pin from the 5 V buck; bench-verify
F4	High	Gigabit uses all 4 tether pairs	No conductors for hardwired surface E-stop	100BASE-TX + E-stop loop on pairs 4/5, 7/8
F5	Medium	Contactor inrush ≈ 1338 A theoretical	Contact welding → E-stop failure	10 Ω / 5 W pre-charge, 200 ms arm delay
F6	Medium	Arming/E-stop switches in main current path (if wired that way)	Need 80+ A DC-breaking switches	Put all switches in the coil loop
F7	Medium	Voltage-only cutoff under 1.5 V sag	False cutoffs / missed depletion	Sag-compensated V_{oc} + INA226 coulomb counting
F8	Medium	Shunt dissipates 3.2 W at 80 A	PDB hot spot	≥ 5 W Kelvin shunt, copper bus bar
F9	Low	LED lamps run in air	Lamp overheating on deck	Depth-gated 20 % limit
F10	Low	Magnetometer near high-current wiring	Heading error	Twist ESC leads, ≥ 10 cm separation, calibrate with thrusters running
F11	Low	Hub drop into UNO Q	Brown-out resets at peak load	Set 5 V buck to 5.1–5.2 V; measure at board
F12	Low	ESC regen spikes near 26 V rating	ESC failure on reversal	Command slew limit, TVS + bulk cap on ESC bus

15. Verification test plan

Every assumed value in this report has a test that replaces it with a measurement. Run in this order; update calc inputs and re-issue the report.

#	Test	Method	Pass criterion
T1	Pack internal resistance	Step 0 → 20 A load, ΔV/ΔI at 100 ms	≤ 20 mΩ pack (≤ 3.3 mΩ/cell)
T2	Hotel load	INA226 log in C1 for 10 min, camera streaming	5 V rail ≤ 7 W typical
T3	UNO Q host mode	Power from 5 V buck via hub; lsusb shows hub, NIC, camera	All devices enumerate after 20 cold boots
T4	Thruster curve	One thruster in a tank at 20/40/60/80/100 % command at 22 V	Measured P within ±15 % of $P_{cap}\cdot u^3$; refine exponent
T5	Power cap	Full command at 25.2 V	≤ 420 W per thruster
T6	All-thrusters burst	C6 for 10 s at nominal, logged	Bus ≥ 20 V, no fuse trip, ESC case ≤ 70 °C
T7	Contactor + pre-charge	Scope bus voltage and coil current on arm	No inrush > 10 A; arm in < 300 ms
T8	E-stop timing	Press surface stop at C5; scope ESC bus	Bus < 5 V within 100 ms
T9	Tether loop	Measure loop resistance and sense current through 50 m	≤ 4 Ω; ≥ 15 mA
T10	Thermal soak	C4 for 60 min in tank; log hull temp	Hull air ≤ 50 °C
T11	Heartbeat loss	Unplug Ethernet at C5	Thrusters neutral ≤ 500 ms
T12	Leak response	Wet one probe	C7 engaged ≤ 200 ms, surface alarm

Appendix A. Formulas used

$P = V \cdot I$; $I = P / V$; $E = P \cdot t$; $Q = I \cdot t$

Converter input: $P_{in} = P_{out} / \eta$; $I_{in} = P_{out} / (\eta \cdot V_{in})$

Battery current with sag: $I = [V_{oc} - \sqrt{(V_{oc}^2 - 4 \cdot R \cdot P)}] / (2R)$; $V_{bus} = V_{oc} - I \cdot R$

Conduction loss: $P = I^2 \cdot R$

Thruster: $P_{full}(V) = 645 \cdot (V/20)^{2.24}$; $P(u) = P_{max} \cdot u^3$; thrust $T \propto P^{2/3}$

Throttle cap: $u_{max} = (P_{cap} / P_{full}(V))^{1/3}$

Runtime: $t = Q_{usable} / I_{avg}$

RC pre-charge: $\tau = R \cdot C$; $E_R = \frac{1}{2} \cdot C \cdot V^2$

IPC-2221 external trace: $I = 0.048 \cdot \Delta T^{0.44} \cdot A^{0.725}$ (A in mil²)

INA226: CAL = 0.00512 / (Current_LSB · R_{shunt}); Power_LSB = 25 · Current_LSB

Appendix B. Data sources and confidence

Item	Source	Confidence
T200 thruster 12/16/20 V points, 7–20 V rating	Blue Robotics product specifications	High
Basic ESC ≈30 A continuous, 26 V	Blue Robotics forum guidance	Medium
UNO Q 5 V / 3 A USB-C, 7–24 V VIN; USB host disabled on VIN	Arduino datasheet / user manual and Arduino forum reports	High / Medium
BNO055, MS5837, BME280, INA226 currents	Manufacturer datasheets (typical values)	High
UNO Q, hub, camera consumption	Typical class values — to be replaced by T2	Medium
Battery IR, C-rating	Typical 6S 8 Ah pack — to be replaced by T1	Medium
ESC idle current, capacitance	Typical BLHeli-class — to be replaced by T6/T7	Low–Medium
Contactor coil	Typical 24 V, 3 W — confirm against chosen part	Medium

All calculations were generated from a single parametric model; changing any input (thruster model, battery IR, cap, budget) regenerates every table consistently.